barrier, this probability is quite small until the kinetic energy is close to the barrier maximum.

As mentioned in §10.3, in 1919 Rutherford caused artificial transmutation of one element into another by using the α-particles emitted in the radioactive decay of ^{214}Po as projectiles. These α-particles had sufficient kinetic energy upon emission from the polonium nucleus (7.7 MeV) to overcome the Coulomb barrier and react with nitrogen nuclei. However, the kinetic energy of α-particles emitted in radioactive decay is generally insufficient to overcome the Coulomb barrier to react with nuclei of higher atomic numbers. Consequently, means had to be devised for acceleration of the charged projectiles to kinetic energies sufficient to achieve reaction.

The principle to be used to achieve the higher kinetic energies was obvious. The projectile particles would have to be ionized to obtain positively charged ions. If these ions could be accelerated through a potential difference of 1000 V, they would acquire 1000 eV additional kinetic energy, per unit of charge. If an α-particle of a +2 charge was to be accelerated through a potential difference of 10^6 V, it would acquire an additional kinetic energy or 2 MeV (eqn. (3.6)). The problem of obtaining the desired kinetic energy involved two aspects: first, the production of the charged particles: second, the acceleration through the necessary potential difference.

16.2. ION SOURCE

The problem of producing the positive ions is in principle relatively simple. If a gas is bombarded by energetic electrons, the atoms of the gas are ionized and positive ions produced. Figure 16.1 shows a simple ion source.

As hydrogen gas flows into the region above the filament, the electrons being emitted by the filament are accelerated to an anode (a typical voltage drop over the electrodes B_1—B_2 may be 100 V) and in their passage through the gas cause ionization. The positive ions are extracted by attraction to a negative electrode (the voltage drop over S_1—S_2 may be 1—10 kV) into the accelerator region. The vacuum at the beam extraction is of the order 10^{-4} Pa but in the ionization compartment it is normally 10^{-2} Pa. The basic principle, bombardment with electrons, is similar in the different types of ion sources that have been developed to meet the demands of the variety of ions and accelerator types in use. High ion currents of all elements in a variety of charge states can now be produced.

16.3. SINGLE-STAGE ACCELERATORS

The first successful accelerator for causing nuclear transmutations was developed by Cockcroft and Walton in 1932 and used a high voltage across an acceleration space.

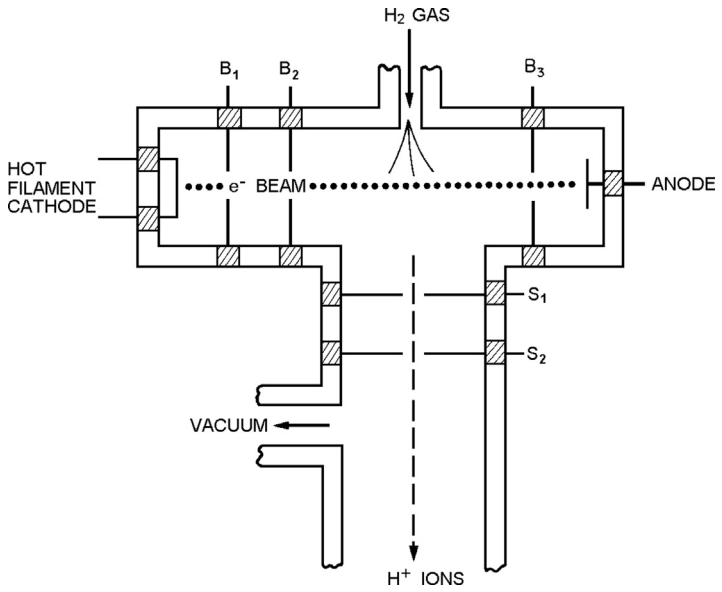


Figure 16.1 A schematic drawing of an ion source producing protons by electron bombardment of hydrogen gas.

Gamow's development of the tunnelling theory for α-decay led Houtermans to suggest an opposite reaction, a high energy proton tunnelling through the Coulomb barrier into a target nucleus. Calculations showed that such tunnelling could occur even at about 100 keV. Cockcroft, who was an electrical engineer, realized that it would be possible to build a high tension generator capable of accelerating protons to such energies (although even Rutherford first thought it technically impossible).

Cockcroft-Walton accelerators are still used for obtaining "low" kinetic energies, up to 4 MeV for protons. Presently, such accelerators are used in many installations as the first stage of acceleration, *injector*, in a more complex machine designed to produce high energy beams (see Figures 16.8 and 16.9).

In a single-stage accelerator, such as a Cockcroft-Walton type, the total potential produced from a high voltage generator is imposed across the accelerator, i.e. between the ion source and the target. The kinetic energy, E_{kin}, of the projectile is:

$$E_{\text{kin}} = nqV \tag{16.1}$$

where q is the charge of the accelerated ions (Coulomb) and V is the imposed potential across the acceleration gap. This equation is identical to (3.6) except that n is the number of accelerating stages ($n = 1$ for the Cockcroft-Walton machine).

In recent years small and relatively inexpensive accelerators have come into use based on the Cockcroft-Walton principle. These are known as *transformer-rectifier accelerators* and are primarily used for acceleration of electrons or acceleration of deuterons for production of neutrons through the reaction:

$$ {}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} \tag{16.2} $$

Tritium targets are bombarded by accelerated deuterons. Tunneling of the Coulomb barrier (see Fig. 16.10) results in a good yield for this reaction even for energies of 0.1 MeV.

Figure 16.2 illustrates an inexpensive transformer-rectifier type accelerator. Deuterium molecules leak through a heated palladium foil into the vacuum of the ion source, where a high frequency electric field decomposes the deuterium molecules to form a plasma of D^{+1} ions and electrons. The deuterium ions are extracted from the ion source with a relatively low negative potential to enter the acceleration tube with approximately 2.5 keV kinetic energy. The voltage difference of 100 kV across the acceleration space is obtained from a transformer and rectifier unit coupled to a set of cylindrical electrodes connected by a resistor chain.

The cylindrical electrodes serve the purpose both of acceleration and of focusing the ion beam into the proper path. As the beam particles exit from the last electrode they drift through a short tube and strike the target, which consists of a metal foil covered with titanium in which tritium has been absorbed. Approximately 200 GBq of tritium

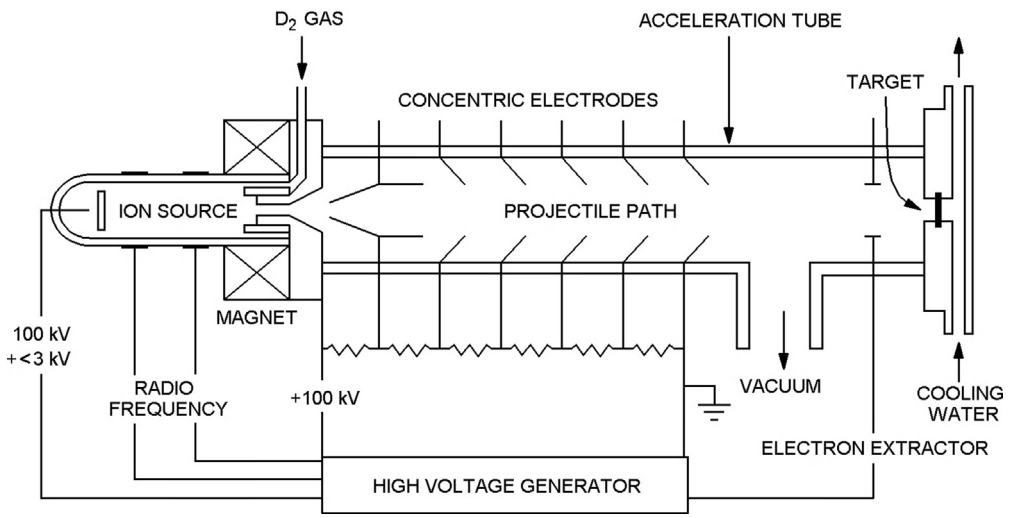


Figure 16.2 Principle of a single-stage linear accelerator. The D_2 gas enters the vacuum through a palladium membrane.

can be absorbed as titanium tritide on an area of 5 cm^2. The target is cooled by water to minimize tritium evaporation. The passage of the beam through the target also results in the emission of electrons which are removed by attraction to an anode to avoid interference with the beam.

With a voltage of 100 kV and an ion current of 0.5 mA, this accelerator can produce approximately 10^{10} n s^{-1} with an energy of 14 MeV. It is often desirable to reduce the kinetic energy of these neutrons to thermal values since thermal neutrons have a much larger probability of interacting with target nuclei. Moderation of the neutrons to thermal energies ($\approx$0.025 eV) can be accomplished by placing water or paraffin around the target (see §§7.6 and 19.5). The flux of the thermal neutrons is relatively low, on the order of 10^8 n cm^{-2} s^{-1}. However, the production rate of neutrons increases as the beam energy and beam current increases.

Small accelerators of similar design are commercially available with such dimensions that they can be lowered into bore-holes for in situ neutron activation analysis during gas or oil prospecting.

More advanced designs of this type of accelerator use coupling of transformers, rectifiers, and condensers to increase the voltage. With the larger single-stage accelerators it is possible to obtain beams of protons and deuterons which have energies of several MeV and currents of about 10 mA.

16.4. VAN DE GRAAFF ACCELERATORS

A type of electrostatic accelerator that has been of great value to nuclear science since its invention is that designed by van de Graaff in 1931. The van de Graaff accelerator (VdG) can provide beams of higher energy than the single-stage Cockcroft-Walton accelerators with a very precisely defined energy. In a modification known as the tandem-VdG beams of 20 MeV protons and 30 MeV α-particles can be achieved. Positive ions of higher Z and electrons can also be accelerated in VdG machines.

The principle of the VdG accelerator is illustrated in Figure 16.3. A rapidly moving belt of a non-conducting material such as rubberized fabric accumulates positive charge as it passes an array of sharp spray points which transfer electrons from the belt to the spray points. The positive charge on the belt is continuously transferred by the movement of the belt away from ground. At the high-voltage terminal, a hollow metal sphere or nearly cylindrical shape, another set of spray points neutralize the charges on the belt by electrons emitted from the spray points. The result is the transfer of the positive charge to the sphere. By a continuous process of transfer of positive charge to the sphere via movement of the belt a high potential relative to ground can be built up on the sphere. The limit of the voltage that can be accumulated on the hollow electrode is determined by its discharge potential to the surrounding housing. If it is insulated by some pressurized non-conducting gas such as N_2, CO_2, or SF_6, potentials of $\sim$15 MV can be